Improving Ground Deformation Classification by Integrating InSAR Time Series With Geospatial Information

Yingbo Dong, Lorenzo Nava, Riccardo Palamà, Oriol Monserrat, Davide Festa, Mario Floris, Ascanio Rosi, and Filippo Catani

***Abstract*—Classifying ground deformation processes, such as landslides, subsidence, deep-seated gravitational slope deformations (DSGSDs), and mining-induced deformations, is key for large-scale hazard assessment and national land-use management. Earth observation provides heterogeneous data over the same geographic region, including interferometric synthetic aperture radar (InSAR) time series (TS), multispectral imagery, and terrain products. However, effectively integrating such spatiotemporal information from multimodal datasets remains a major challenge. To fully use the rich information contained in the TS and to exploit the complementary strengths of spatial and temporal data, we propose a dual-branch deep learning (DL) approach that integrates InSAR ground deformation TS with geospatial information for classifying slow-moving ground deformation processes. To validate the approach, we construct a ground deformation dataset containing over 26 000 active deformation areas (ADAs), labeled into four deformation types: landslide, subsidence, DSGSD, and mining. Results demonstrate that our model achieves an overall classification accuracy exceeding 90% on both ascending and descending test datasets, though confusion remains between certain classes, such as landslides and DSGSD. Explainable AI (XAI) analysis indicates that spatial and morphological features contribute more significantly to classification performance than temporal deformation patterns, with clearer distinctions for subsidence and mining, but more overlap between landslides and DSGSDs. This work highlights the strength of multimodal data fusion method to classify ground deformation processes, while setting the stage for future research.**

***Index Terms*—Deep learning (DL), ground deformation classification, interferometric synthetic aperture radar (InSAR), multimodal data fusion.**

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I. Introduction

GROUND deformation processes, such as landslides, subsidence, deep-seated gravitational slope deformations (DSGSDs), and anthropogenic deformations such as mining-induced subsidence and uplift, or settlement caused by construction activities, can have severe impacts on infrastructure and human safety [1], [2], [3], for example, the Ca' Lita landslide in the Northern Apennines of Italy, reactivated in 2002 and 2004, advanced approximately 400 m, threatening nearby villages and critical infrastructure [4]. Classifying these processes is key to tailor appropriate mitigation strategies and ensuring timely risk management, as different processes require different approaches.

Traditionally, the identification of these deformations has primarily relied on geomorphological interpretation, which suffers from low efficiency and limited real-time performance, while also demanding substantial workpower and material resources [5]. At scale, the rapid development and integration of geodetic tools, satellite remote sensing (RS), geographic information systems, and digital photogrammetry have provided technical support for real-time and continuous monitoring. Among these technologies, interferometric synthetic aperture radar (InSAR) has enabled large-scale spatial and temporal monitoring, allowing for all-weather observation of ground deformation [6], [7], [8]. Furthermore, advanced differential InSAR techniques (A-DInSAR) such as persistent scatterer interferometry (PSI) [9], [10] and interferogram stacking [11], [12] leverage long time series (TS) of SAR data to achieve millimeter-level measurement accuracy. PSI enables the detection of potential persistent scatterers (PSs), facilitating detailed monitoring of ground deformation along the radar line of sight (LOS).

Although InSAR effectively captures various ground deformation processes, few approaches have been developed to automatically distinguishing between them. Existing methods can be broadly divided into two categories. The first category focuses on classifying ground deformation TS trends (e.g., stable, linear, quadratic, or seasonal) [13], using either traditional

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machine learning (ML) or deep learning (DL) techniques. ML-based approaches typically rely on handcrafted features such as statistical indicators, trend coefficients, and annual mean velocity, combined with classifiers such as random forest, K-nearest neighbors, or XGBoost [14], [15]. For example, statistical and distribution-based features such as variance, skewness, and kurtosis have been commonly used to represent deformation TS, enabling efficient classification with models such as random forest and XGBoost [16]. More recently, DL models such as recurrent neural networks (RNNs), long short-term memory (LSTM) [17], and convolutional neural networks (CNNs) have been used to learn directly from raw TS data and identify trend patterns [18], [19], [20], [21]. However, these models generally aim to classify deformation behaviors rather than the underlying geophysical processes responsible for them.

The second category attempts to classify ground deformation processes. Early efforts often relied on manually defining deformation events [14], or using rule-based classifiers such as PSI target characterization approach [22], [23], as well as decision trees [24] for deformational processes classification from InSAR TS. However, these methods have limited generalization and are typically applicable to specific regions. Recent studies have further automated the classification process by applying unsupervised learning techniques. For instance, [25] combined PCA and K-means clustering to detect spatially and temporally coherent deformation phenomena from fused ascending/descending Sentinel-1 InSAR TS, identifying multiple deformation patterns associated with landslides and anthropogenic activities. More recent studies have increasingly relied on supervised ML algorithms for automatic classification of deformation types [26], [27]. For example, an oriented R-CNN was developed to detect and classify subsidence using InSAR imagery and multisource data [28].

While these methods contribute to classify different types of deformation processes, they typically rely on manually derived features and do not fully leverage the rich temporal and spatial morphological information available from TS and geospatial datasets. This highlights the need for more comprehensive approaches that integrate multiple data sources, e.g., InSAR ground deformation TS, optical RS images, the digital elevation model (DEM), and leverage their spatiotemporal information. Recent studies have explored multimodal data fusion techniques to integrate TS and image data. For example, Benedetti et al. [29] proposed a DL framework that integrates multitemporal Sentinel-2 data and very high-resolution images for land cover classification; Hayat et al. [30] proposed an LSTM-based fusion module for fusing TS and image data in healthcare applications; and Rasekh et al. [31] developed an attention-based dual-encoder model to fuse TS and image data for clinical prediction tasks.

Inspired by how geomorphologists interpret deformation through both temporal evolution and spatial cues from topography and imagery, we propose a multimodal fusion approach that integrates TS and geospatial features for classifying ground deformation processes (i.e., landslide, subsidence, mining, and DSGSD). To achieve this, we develop a dual-branch DL model that processes TS data using a hybrid 1-D CNN–LSTM network and spatial data using a 2-D CNN. The two feature streams are then combined through a data fusion module. The key contribution of this study lies in introducing multimodal DL to the classification of ground deformation processes, effectively leveraging both temporal and spatial information. Experiments conducted on a test dataset from Italy demonstrate that our approach significantly outperforms single-modality baselines. Furthermore, explainable AI (XAI) techniques are used to interpret the model's predictions and provide insights into the decision-making process.

The rest of this article is organized as follows. Section II details the study area and materials, Section III describes the dataset creation, model construction, and evaluation, Section IV presents the experiments results and discussions, followed by the conclusion in Section V.

II. Study Area and Materials

A. Study Area

The study focuses on Italy, encompassing a variety of geological and environmental settings. Italy is known for its diverse topography, ranging from the mountainous regions of the Alps and Apennines to the flat plains of the Po Valley. This varied landscape is associated with numerous geohazards, including landslides, subsidence, and seismic activity [32]. Moreover, human-induced deformation processes such as mining, oil and gas extraction, and geothermal energy exploitation are also widespread [33]. This makes Italy an ideal natural laboratory for evaluating the performance of classification techniques across varied deformation types.

B. Materials

1) EGMS Data: This study uses InSAR ground displacements TS data provided by the European Ground Motion Service (EGMS) [34]. The EGMS Level-2b dataset, also referred to as the calibrated product, includes LOS velocity maps in both ascending and descending orbits. The calibrated product is referenced to a global navigation satellite system (GNSS)-derived model, offering absolute displacement measurements, which allows for consistent comparisons across different regions. The dataset we used spans from 2015 to 2021, and the temporal resolution is 6 days. Each measurement point is geolocated and includes detailed quality metrics, ensuring high reliability for monitoring ground deformation patterns. Using this dataset, the study benefits from the high precision and stability of the EGMS products, which have been validated through extensive testing and benchmarking [35], [36].

2) Geospatial Data: To complement the InSAR data, various geospatial variables were used. Sentinel-2 imagery, with a spatial resolution of 10 m, was used across four spectral bands: red, green, blue (RGB), and near-infrared (NIR). The Sentinel-2 imagery used in this study was taken in September 2021, a period chosen for its minimal cloud cover. This timing also matches the later years of the EGMS dataset. The DEM used in this study is the EU-DEM, which covers Europe at a

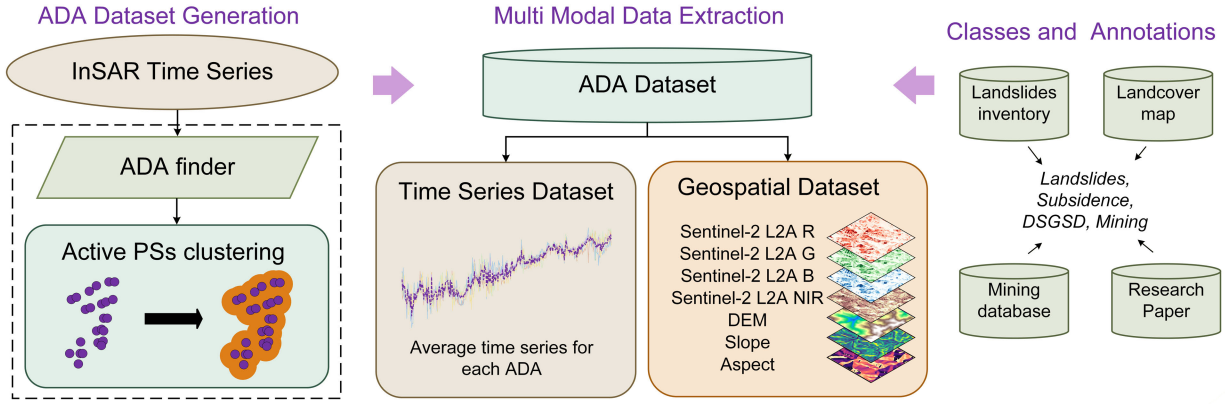


Fig. 1. Workflow of dataset creation. The workflow begins with the identification of ADAs from EGMS ground displacement TS using the ADA finder tool. For each ADA, InSAR TS and geospatial datasets are extracted, including Sentinel-2 spectral bands, DEM, slope, and aspect. Part of these data are labeled using some reference like landslides inventories and land cover maps.

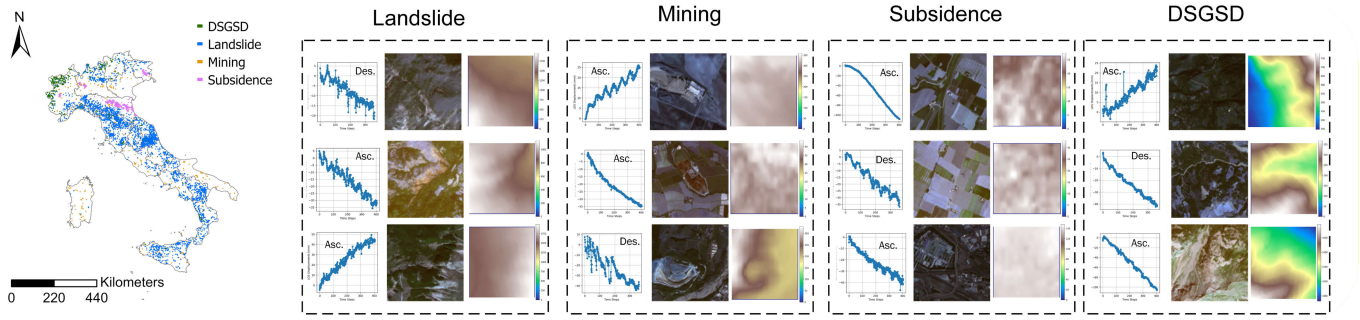


Fig. 2. Spatial distribution and examples of our labeled multimodal ground deformation dataset. There are four different ground deformation types in our dataset, and for every ADA, we extract ground displacements TS and Sentinel-2 images and DEM.

resolution of 30 m and was used to extract key topographic parameters such as slope and aspect.

III. Methodology

In this study, we used the Active Deformation Areas (ADAs) dataset illustrated in [27], where each polygon was labeled according to ground deformation types (see Fig. 1). For each ADA, we extracted TS and spatial data to build the input dataset. Based on the dataset, we then designed a DL model capable of combining temporal and spatial features. Finally, the model was evaluated using standard evaluation metrics.

A. Dataset Creation

1) ADAs Dataset: Due to the vast volume of InSAR data, accurately identifying ADAs is essential for focusing analysis on relevant regions and improving overall efficiency. We used an ADAs dataset generated by ADAfinder [37], an automated tool designed for the detection of ADAs based on PSI TS clustering. Full description of the methodology for ADAs detection can be found in [38].

2) MultiModal Data Extraction: For each identified ADA, PS points from both ascending and descending orbits located within the ADA polygon were extracted, and their average LOS displacements' TS was calculated to represent the temporal deformation pattern of the area. To incorporate spatial information, the centroid of each ADA polygon was determined, and a square bounding box with a side length of 1280 m was created around it. Image datasets, including Sentinel-2 L2A imagery and DEM derived layers, were extracted within this bounding box. All the images were resized to a fixed size of 128 $\times$ 128 pixels to ensure consistency and compatibility with the model input.

3) Classes and Annotations: In all, 17 795 ADAs were labeled into four classes: landslide, mining, subsidence, and DSGSD. These categories were selected because they represent slow deformation processes that can be effectively monitored by satellite. Landslides and DSGSD were annotated with assistance from the Italian Landslide Inventory (IFFI), which provides extensive documentation of landslide activities across Italy [39]. Subsidence was primarily identified in the Po River Basin in Emilia-Romagna, and the dataset was derived from a subsidence map [40]. Mining-induced deformations were mainly annotated by the ISPRA Italy mining site database, with additional insights provided by the results in [41]. Fig. 2. shows the spatial distribution and examples from the dataset.

4) Dataset Splits: The ascending dataset contains a total of 13 499 ADAs; while the descending contains 13 528 ADAs. To ensure consistent data distribution and avoid overlap, the dataset was divided 80% for training and 20% for testing. The

TABLE I
NUMBER OF PATCHES FOR TRAINING VALIDATION AND TEST SETS

	Dataset	Training	Test	Total
Asc.	Landslide	5394	1349	6743
	Mining	713	178	891
	Subsidence	3547	887	4434
	DSGSD	1145	286	1431
	Total	10799	2700	13499
Des.	Landslide	5043	1261	6304
	Mining	712	178	890
	Subsidence	3725	931	4656
	DSGSD	1342	336	1678
	Total	10822	2706	13528

distribution of samples for each class and set is presented in Table I.

B. Model Design

In this study, we propose a novel feature-level multimodal data fusion approach that integrates both TS and geospatial image (GI) data from scratch. This method leverages DL techniques to capture and analyze complex relationships within the dataset.

A schematic illustration of the proposed model is shown in Fig. 3. The input of our DL model consists of N labeled multimodal samples, denoted as $\{x^i, y^i\}_{i=1}^N$, where each sample $x^i = \{x_{\text{ts}}^i, x_{\text{gi}}^i\}$ includes a TS component $x_{\text{ts}}^i \in \mathbb{R}^{T\times 1}$ and a GI component $x_{\text{gi}}^i \in \mathbb{R}^{H\times W\times D}$, and $y^i \in \{0, 1, \ldots, C-1\}$ is the corresponding class label. Here, N is the total number of training samples, T is the TS length, H, W, and D denote the height, width, and number of channels of the GI, respectively, and C is the total number of deformation classes. Notably, our model adopts a dual-branch architecture.

1) *TS Processing Branch:* A hybrid 1-D CNN–LSTM network, which applies to capture temporal dependencies and multiscale patterns in TS data.
2) *GI Processing Branch:* A three-layer 2-D CNN designed to extract spatial features from GI data.

After feature extraction, the outputs from both the branches are fused using an LSTM-based feature fusion module. Finally, a fully connected softmax layer is applied for classification. The backbone networks for both the branches were chosen based on preliminary comparisons and literature evidence (described in Section IV).

1) TS Processing Module: Given an input TS x_{ts}^i, the model first applies three stacked 1-D convolutional layers (Conv1D) with different kernel sizes (2, 5, 5) and strides (1, 3, 5) to extract multiscale TS features. Each convolutional layer operates as follows:

$$F_{\text{ts}}^l = \sigma\left(W^l * F_{\text{ts}}^{l-1} + b^l\right) \tag{1}$$

where F_{ts}^l represents the feature maps at layer l, W^l is the convolution kernel, $*$ denotes the convolution operation, b^l is the bias term, and σ represents the rectifier linear unit (ReLU) activation function. The initial input is set as $F_{\text{ts}}^0 = x_{\text{ts}}^i$.

After convolution, an average pooling layer is applied to reduce the temporal resolution while preserving the key patterns [42]

$$P = \frac{1}{k}\sum_{j=1}^{k}\left(F_{\text{ts}}^l\right)_j \tag{2}$$

where k is the pooling window size, and $(F_{\text{ts}}^l)_j$ is the feature vector at the jth time step of the lth layer output.

A dropout layer with a rate of 0.2 is added to mitigate overfitting

$$F_{\text{drop}} = \text{Dropout}\,(P, 0.2)\,. \tag{3}$$

The TS features F_{drop} are then fed into an LSTM layer with 112 hidden units [43] and then get every hidden state of the LSTM layer

$$h_T = \text{LSTM}\left(F_{\text{drop}}\right). \tag{4}$$

LSTM incorporates a gating mechanism to address the vanishing/exploding gradient problems, enabling RNNs to learn long-term dependencies. As shown in Fig. 4, LSTM introduces memory cells and three gates to control the flow of information. The operations of an LSTM unit at time step t can be described as follows:

Forget gate determines which information from the previous cell state should be discarded

$$f_t = \sigma\left(W_f \cdot [h_{t-1}, x_t] + b_f\right) \tag{5}$$

where f_t is the forget gate activation, h_{t-1} is the previous hidden state, x_t is the current input, and W_f and b_f are the forget gate parameters.

Input gate decides which new information should be added to the cell state

$$i_t = \sigma\left(W_i \cdot [h_{t-1}, x_t] + b_i\right) \tag{6}$$

$$\tilde{C}_t = \tanh\left(W_C \cdot [h_{t-1}, x_t] + b_C\right) \tag{7}$$

where i_t is the input gate activation, and $\tilde{C}_t$ represents candidate values for updating the cell state.

Cell state update combines the forget and input gate decisions to update the memory cell

$$C_t = f_t \otimes C_{t-1} + i_t \otimes \tilde{C}_t \tag{8}$$

where C_t is the new cell state, and $\otimes$ denotes elementwise multiplication.

Output gate determines the new hidden state h_t based on the updated cell state

$$o_t = \sigma\left(W_o \cdot [h_{t-1}, x_t] + b_o\right) \tag{9}$$

$$h_t = o_t \otimes \tanh C_t \tag{10}$$

where o_t is the output gate activation, and h_t is the updated hidden state.

The LSTM output h_T is passed through a flatten layer to generate a 1-D vector

$$F_{\text{flat}}^{\text{ts}} = \text{Flatten}\left(h_T\right). \tag{11}$$

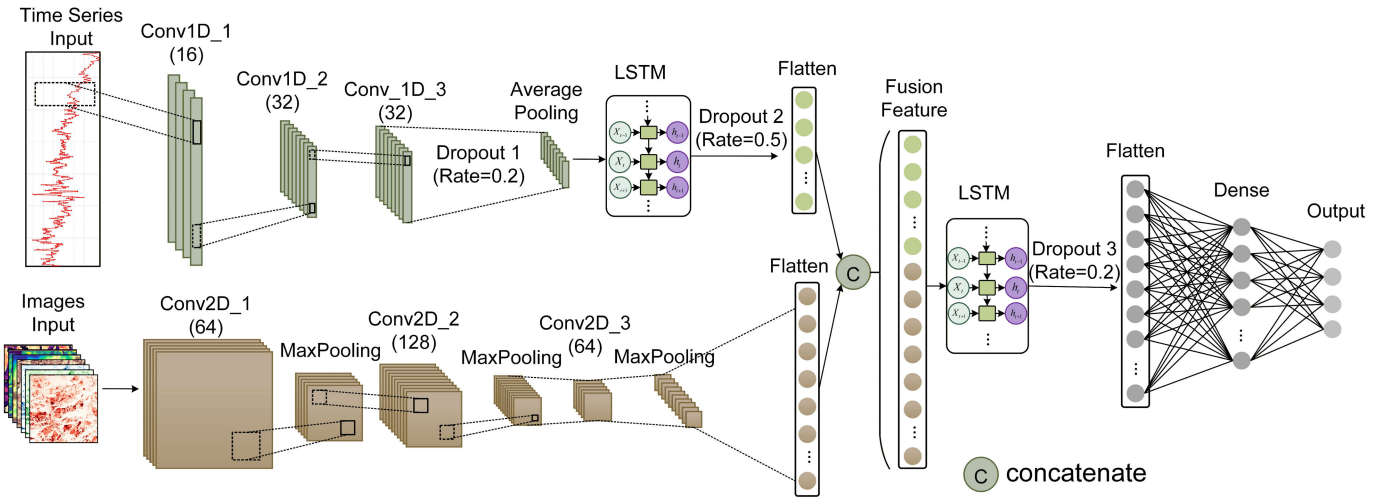


Fig. 3. Framework of the proposed method, which takes seven GIs and InSAR TS as the input.

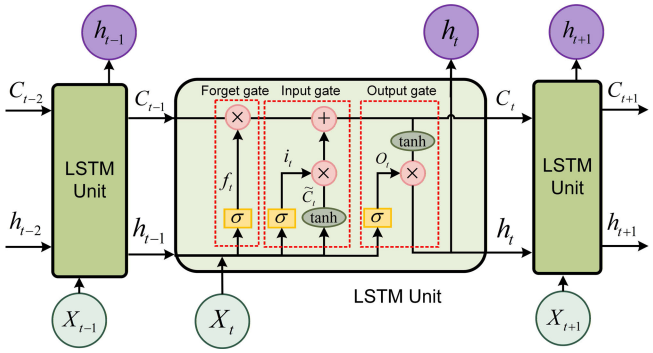


Fig. 4. Schematic of the LSTM unit.

2) Image Processing Module: The GI input x_{gi}^{i} is processed using a three-layer 2-D CNN to extract spatial features. CNNs are well-suited for spatial pattern recognition, as they apply convolutional filters to learn local features and progressively from hierarchical feature representations.

The feature extraction process consists of three convolutional layers (Conv2D), each with a 3×3 kernel size and ReLU activation function, enabling the capture of hierarchical spatial features. Like 1D-CNN, the convolution operation at each layer can be expressed as follows:

$$F_{\text{gi}}^{l} = \sigma\left(W^{l} * F_{\text{gi}}^{l-1} + b^{l}\right). \tag{12}$$

Each convolutional layer is followed by a max-pooling layer with a 2×2 filter size, which downsamples the feature maps while preserving key spatial patterns

$$F_{\text{pool}}^{l} = \max_{(i,j)\,\epsilon R} F^{l}. \tag{13}$$

The output feature maps are then flattened into a 1-D vector

$$F_{\text{flat}}^{\text{gi}} = \text{Flatten}\left(F_{\text{pool}}^{3}\right) \tag{14}$$

where F_{pool}^{3} is the feature map from the last pooling layer.

3) Feature Fusion: The TS features $F_{\text{flat}}^{\text{ts}}$ and GI features $F_{\text{flat}}^{\text{gi}}$ are concatenated to form a fused feature vector f_{fused}.This vector is then processed by an LSTM layer (112 units) to model complex dependencies between TS and GI features.

Ultimately, we selected the LSTM-based fusion strategy because it processes inputs in a sequential manner, allowing the model to learn dependencies across different data modalities over time [44]. The sequential nature of LSTM processing mirrors this decision-making process and facilitates effective integration of temporal and spatial cues in our dataset.

A dropout layer is added to prevent overfitting. Finally, a fully connected softmax layer is applied to produce the final classification output from C classes.

4) Experimental Setting: The experiments were conducted on a workstation equipped with a GeForce RTX 3090 GPU and 24 GB of RAM. The Adam [45] optimizer was used with an initial learning rate of 0.0001, which was gradually adjusted using the ReduceLROnPlateau strategy to enhance model convergence. The network's output is a 1-D tensor of length 4, corresponding to the predicted probabilities for each of the four deformation categories. The training procedure was configured for 100 epochs, with a batch size of 16, constrained by memory limitations. We performed hyperparameter tuning using Keras Tuner with the hyperband strategy, optimizing hyperparameter including the number of convolutional filters, kernel sizes, strides, the number of LSTM units, dropout rates, etc., to achieve the best validation performance. The tuning process converged, and the model parameters described above were obtained accordingly. A tenfold cross-validation strategy was implemented during training to enhance the robustness and reliability of the results. These settings were consistently applied across all the models and comparison methods to ensure uniformity in the experimental setup.

C. Model Evaluation

To evaluate the performance of the developed method, the overall accuracy (OA) and the $F1$ score ($F1$) are used. More

TABLE II

EXPERIMENTAL RESULTS OF DIFFERENT FUSION METHODS ON THE ASCENDING AND DESCENDING DATASETS. WE PRESENT THE $F1$ OF FOUR GROUND DEFORMATION CLASSES AND FOUR OVERALL PERFORMANCE METRICS. VALUES IN PARENTHESES REPRESENT THE STANDARD DEVIATION OVER TENFOLD CROSS-VALIDATION. BOLD VALUES ARE THE BEST

Fusion Method	Dataset	F1(%)				Precision (%)	Recall (%)	mF1 (%)	OA (%)
		Landslide	Mining	Subsidence	DSGSD				
Plain Fusion	Asc.	91.50 (±0.63)	71.67 (±4.33)	98.61 (±0.20)	63.61 (±4.62)	86.27 (±1.2)	78.53 (±3.2)	81.35 (±2.31)	90.20 (±0.83)
	Des.	91.37 (±0.44)	70.78 (±4.21)	98.36 (±0.28)	72.67 (±2.44)	87.38 (±1.04)	80.62 (±2.36)	83.30 (±1.71)	90.54 (±0.62)
Attention Based	Asc.	91.21 (±0.17)	68.58 (±2.04)	98.05 (±0.35)	63.36 (±2.76)	86.05 (±1.48)	76.94 (±1.83)	80.30 (±1.05)	89.70 (±0.22)
	Des.	91.57 (±0.38)	72.23 (±1.98)	98.52 (±0.22)	75.75 (±1.36)	87.61 (±1.33)	82.34 (±1.53)	84.52 (±1.87)	91.00 (±0.44)
LSTM Based (Ours)	Asc.	**92.82** (±0.59)	**75.37** (±2.19)	**98.61** (±0.15)	**71.90** (±3.53)	**88.58** (±0.96)	**81.89** (±1.94)	**84.68** (±1.46)	**91.73** (±0.65)
	Des.	**92.69** (±0.24)	**77.08** (±1.28)	**98.72** (±0.19)	**78.94** (±1.86)	**89.54** (±0.46)	**84.79** (±0.75)	**86.86** (±0.49)	**92.23** (±0.27)

specifically, the mean $F1$ ($mF1$) was computed for four ground deformation classes. Their formulas are defined as follows:

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}} \tag{15}$$

where TP is the true positive which is the number of correctly predicted positive samples, FP is the false positive which is the number of negative samples incorrectly classified as positive, FN is the false negative which is the number of positive samples incorrectly classified as negative, and TN is the true negative which is the number of correctly predicted negative samples

$$mF1 = \frac{1}{C}\sum_{i}^{C} F1_i \tag{16}$$

where C is the number of classes, and for each class i is computed as follows:

$$F1_i = 2 \times \frac{\mathrm{Precision}_i \times \mathrm{Recall}_i}{\mathrm{Precision}_i + \mathrm{Recall}_i} \tag{17}$$

with

$$\mathrm{Precision}_i = \frac{\mathrm{TP}_i}{\mathrm{TP}_i + \mathrm{FP}_i} \tag{18}$$

$$\mathrm{Recall}_i = \frac{\mathrm{TP}_i}{\mathrm{TP}_i + \mathrm{FN}_i}. \tag{19}$$

IV. RESULTS AND DISCUSSION

A. Data Fusion Method Experimental Results

In this study, we designed multimodal data fusion strategies for integrating temporal and spatial information from InSAR ground displacements TS and geospatial data. As part of the comparison and further verification of the proposed approach, we tested several fusion models, including the LSTM-based fusion model [30], as well as two additional feature-level data fusion approaches, which are plain fusion methods [29] and attention-based fusion methods [31]. The quantitative results of these models on our ascending and descending datasets are presented in Table II.

All the three approaches achieved satisfactory classification performance, with $mF1$ exceeding 80% on both the datasets. This confirms the effectiveness of feature-level fusion in capturing the complementary information of TS and spatial data. Among them, the LSTM-based method performed best. Specifically, on the ascending dataset, the LSTM-based method outperformed the plain fusion method in terms of $F1$ score for landslide, mining, and DSGSD by 1.32%, 3.70%, and 8.29%, respectively. Compared with the attention-based method, $F1$ scores improved for all the four categories, with gains of 1.61%, 6.79%, 0.56%, and 2.53%, respectively, with the highest improvements observed in mining and DSGSD. On the descending dataset, the LSTM-based method achieved higher $F1$ scores than the plain fusion method across all the four categories, with increases of 1.32% (landslide), 6.30% (mining), 0.36% (subsidence), and 6.27% (DSGSD). Overall, the LSTM-based model achieved the highest $mF1$ and OA across both the datasets, with $mF1$ values of 84.68% and 86.86% and OA values of 91.73% and 92.23% for the ascending and descending datasets, respectively.

From a classwise perspective, landslide and subsidence exhibited $F1$ scores above 90%, indicating that these categories are relatively easier to distinguish. In contrast, the classification performance for mining and DSGSD was comparatively lower. The confusion matrices (Fig. 5.) further illustrate these findings. The model's most frequent misclassification was between Landslide and DSGSD, with 84 and 64 cases misclassified in the ascending and descending datasets, respectively. Mining was often confused with landslide, with 36 and 42 cases misclassified in both the datasets. In contrast, subsidence classification remained stable across both the datasets, with minimal misclassification errors.

Our model's $mF1$ score standard deviation on both the ascending and descending datasets are within ±1.5%, For each class, the standard deviations of landslide and subsidence are the lowest, generally staying within ±1%, while the other two classes show slightly higher variability. This overall

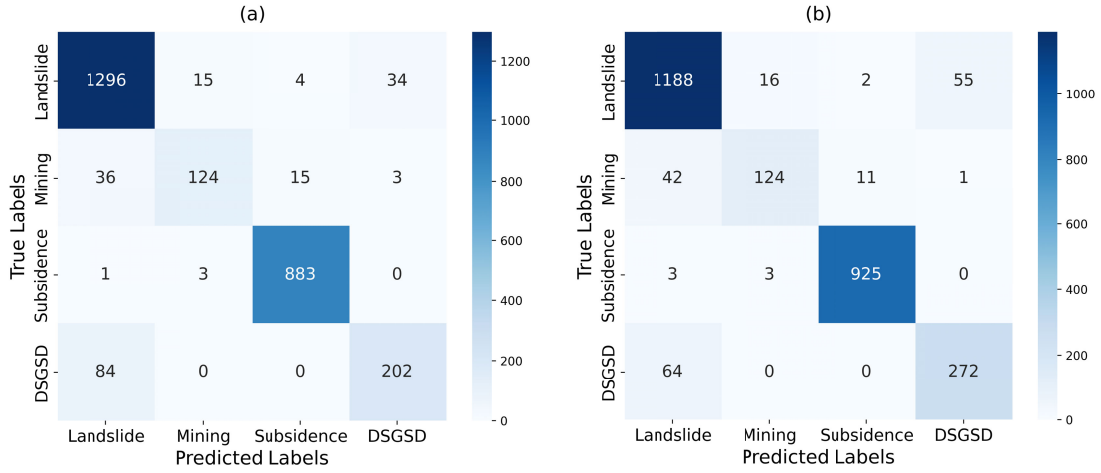


Fig. 5. Confusion matrices of the proposed LSTM-based data fusion model on the test dataset. (a) Ascending dataset. (b) Descending dataset.

TABLE III

EXPERIMENTAL RESULTS ON THE ASCENDING AND DESCENDING DATASETS WITH DIFFERENT INPUT MODALITIES. WE PRESENT THE $F1$ OF FOUR GROUND DEFORMATION CLASSES AND FOUR OVERALL PERFORMANCE METRICS. VALUES IN PARENTHESES REPRESENT THE STANDARD DEVIATION OVER TENFOLD CROSS-VALIDATION. BOLD VALUES ARE THE BEST

Model	Dataset	F1(%)				Precision (%)	Recall (%)	mF1 (%)	OA (%)
		Landslide	Mining	Subsidence	DSGSD				
1D-CNN-LSTM	Asc.	80.74 (±0.41)	30.49 (±6.98)	87.19 (±0.93)	47.34 (±5.61)	68.35 (±1.51)	58.97 (±2.97)	61.44 (±3.28)	77.56 (±0.64)
	Des.	75.63 (±0.77)	22.66 (±5.57)	80.18 (±1.36)	49.86 (±2.67)	68.66 (±0.91)	54.56 (±2.17)	57.08 (±2.51)	72.49 (±1.12)
2D-CNN	Asc.	91.44 (±0.41)	72.06 (±2.58)	98.64 (±0.24)	62.36 (±3.38)	86.62 (±1.06)	77.89 (±1.83)	81.13 (±1.45)	90.15 (±0.56)
	Des.	91.11 (±0.54)	68.50 (±4.69)	98.36 (±0.19)	71.69 (±3.66)	87.24 (±0.73)	79.44 (±3.03)	82.42 (±2.16)	90.23 (±0.73)
Fusion Model	Asc.	**92.82** (±0.59)	**75.37** (±2.19)	**98.61** (±0.15)	**71.90** (±3.53)	**88.58** (±0.96)	**81.89** (±1.94)	**84.68** (±1.46)	**91.73** (±0.65)
	Des.	**92.69** (±0.24)	**77.08** (±1.28)	**98.72** (±0.19)	**78.94** (±1.86)	**89.54** (±0.46)	**84.79** (±0.75)	**86.86** (±0.49)	**92.23** (±0.27)

stability across classes and datasets highlights the reliability and robustness of our model.

B. Ablation Study

To investigate the effectiveness and contribution of each branch in our proposed model, ablation experiments were conducted by training individual unimodal models using only TS or GI and compared their performance with the proposed LSTM-based fused model. As shown in Table III, two ablation experiments were designed based on our fusion scheme. In the first experiment, only the TS processing branch was kept. In contrast, the second experiment only keep images processing branch.

For the ascending dataset, the 1D-CNN–LSTM model achieved the $mF1$ of 61.44%, significantly lower than the 2-D CNN Model (81.13%) and the fusion model (84.68%). This suggests that spatial features contribute considerably to classification accuracy. The fusion model, which integrates both the modalities, achieved the highest OA (91.73%), demonstrating that incorporating both the TS and spatial features leads to improved classification performance.

A similar pattern is observed in the descending dataset result, where the 1D-CNN–LSTM model achieved the $mF1$ of 57.08%, while the 2D-CNN model reached 82.42%. The fusion model further improved performance, achieving the $mF1$ of 86.86% and the OA of 92.23%, indicating that the combined information from both the TS and spatial data provides a more comprehensive representation for classification.

These results highlight the complementary nature of TS and spatial data, confirming that our multimodal fusion approach enhances classification performance compared with unimodal inputs.

C. Classification Results on Unlabeled Data

After validating the trained models using test dataset, we applied the models to classify 99 402 unlabeled ADA instances across Italy. This large-scale application aimed to explore the spatial distribution of predicted deformation types throughout the country.

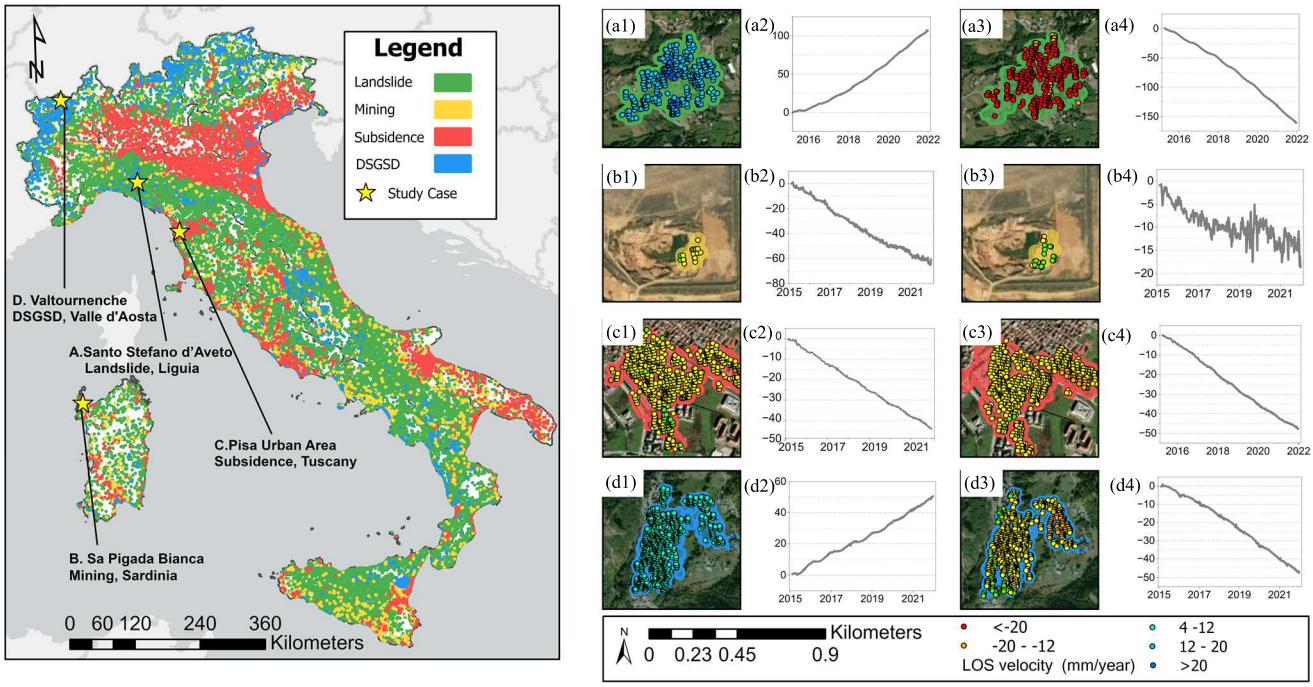


Fig. 6. Left: prediction results on Italy and the geographical locations of the four study cases. Right: detailed visualization of the four study cases, each row represents one case. Labels (a1)–(d1) show the ascending PS points, (a2)–(d2) show the ascending mean LOS displacements (mm), (a3)–(d3) show the descending PS points, and (a4)–(d4) show the descending mean LOS displacements (mm).

The classification results from the LSTM-based fusion model trained on ascending data are visualized in Fig. 6 using a four-color mapping scheme [27], where each ground deformation class (i.e., landslide, mining, subsidence, and DSGSD) is represented by a distinct bright color. To enhance the visibility of the relatively small ADAs regions with the national scale, the boundary width of each ADA polygon was increased, which may lead to partial overlaps in areas where ADAs are densely clustered. The classification yields a total of 71 101 ADAs classified as landslides, 5191 as mining, 18 853 as subsidence, and 4257 as DSGSD.

To better illustrate the classification results, we selected four validated examples for further analysis. As shown in Fig. 6, these cases represent different types of ground deformation and allow for a more concrete evaluation of model behavior. Case **A** is a landslide located in Santo Stefano d'Aveto, Liguria, validated by the IFFI inventory. Case **B** is a mining site located in Sa Pigada Bianca, Sardinia, validated by [46]. Case **C** is a subsidence area located in Pisa, Tuscany region, validated by [47]. Case **D** is a DSGSD case located in Valle d'Aosta, validated by IFFI inventory. Among these examples, both the landslide (Case **A**) and DSGSD (Case **D**) exhibit an increasing trend in ascending data and a decreasing trend in descending data. The mining (Case **B**) and subsidence (Case **C**) cases show decreasing trends in both the ascending and descending data. These variations in TS patterns are closely related to terrain factors such as slope and aspect.

These observations align with the geometric principles of InSAR. As shown in Table IV, Cases **A** and **D** located in mountainous regions with higher elevations (1006 and 1566 m, respectively) and steeper slopes (7° and 14°) suggest dominant horizontal slide motion (e.g., eastward or slope-parallel movement). Cases **B** and **C** are situated in low-lying, nearly flat regions (27 and 4 m elevation; 1° slope), where deformation is primarily vertical.

TABLE IV
GEOMORPHOLOGICAL PARAMETERS OF FOUR REPRESENTATIVE ADAS

Case	Predicted Class	Mean elevation (m)	Mean slope (°)	Mean aspect (°)	Area (m^2)
A	Landslide	1006	7	273	137,957
B	Mining	27	1	113	9,416
C	Subsidence	4	1	174	201,513
D	DSGSD	1566	14	245	311,071

D. Interpretation of Model Decision-Making

To further analyze the interpretability of our proposed model, we used GradientExplainer [48] to calculate Shapley additive explanations (SHAP) values. For visualization, we separated the SHAP values of both the TS and image branches, visualizing the heatmaps for different categories. These heatmaps help us identify key region influencing the model's decisions.

As shown in Fig. 7, the left columns display the SHAP heat maps for the image branch. In these maps, red pixels increase the probability of the image being classified as the corresponding class while blue pixels decrease it. The sum of the SHAP values for a given input equals the difference between the model's prediction for that input and its

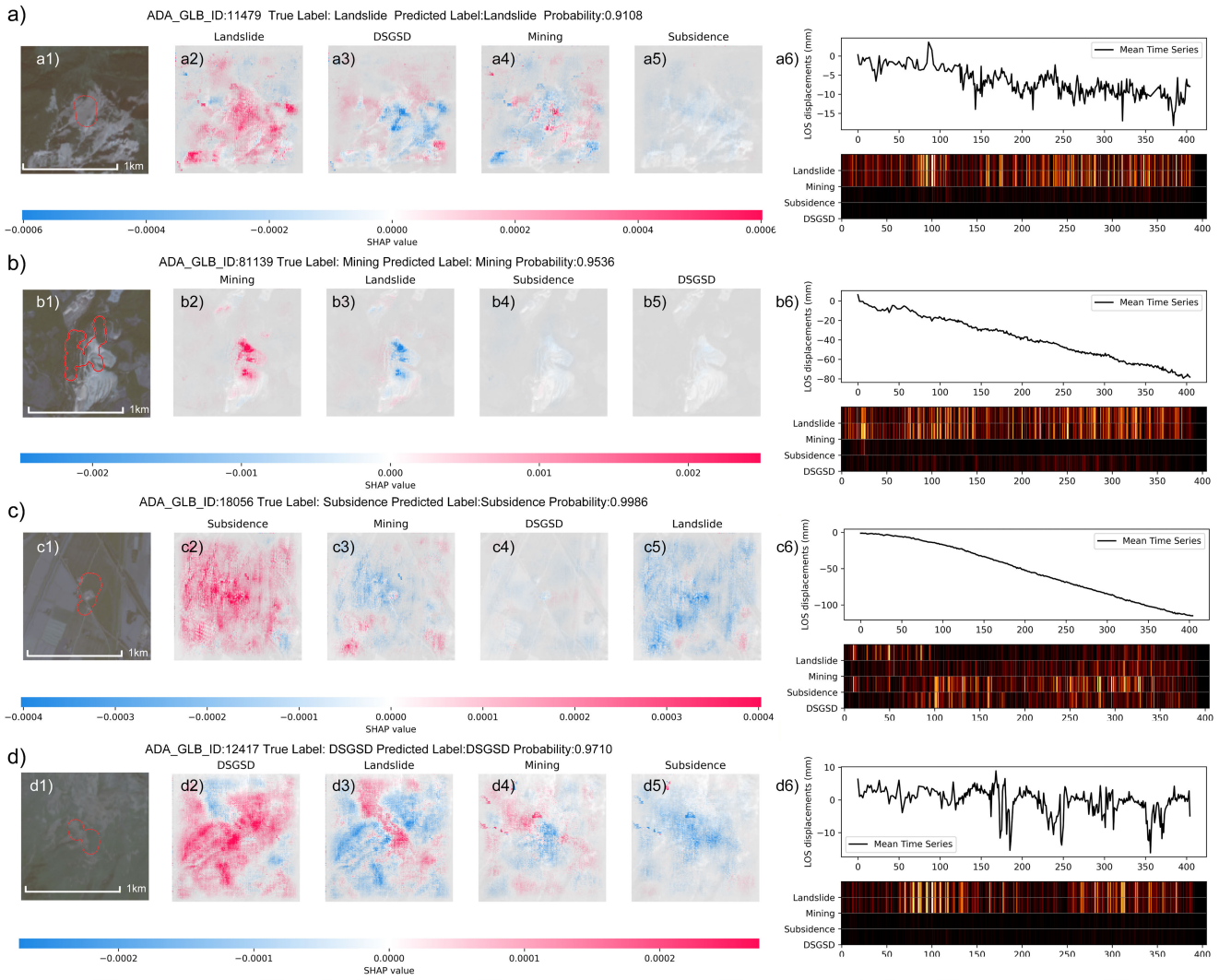


Fig. 7. SHAP visualization heat map examples. (a): (1) RGB image of test example, (2)–(5) image branch SHAP heat map for four classes from the proposed model, and (6) mean ground displacements TS and absolute SHAP values heat map for four classes from the proposed model. (b)–(d) Organized in the same way.

average prediction across a background dataset. Fig. 8. shows the summary plot generated of the image branch using 100 correctly predicted samples for each class in the test set. It shows the feature importance ranking and how they impact the model's output. Overall, topographic bands including DEM, slope, and aspect appear among the most important features across all the four classes. However, our model does not rely solely on single feature values but rather captures the spatial information embedded in the image's patches. For instance, in landslide and DSGSD cases, SHAP values are more localized, which corresponds to the fact that landslides usually affect specific slope sections rather than the entire region, whereas in mining areas, SHAP contributions are concentrated in regions undergoing excavation activities. In contrast, subsidence areas exhibit a more uniform SHAP distribution across the entire region due to the relatively flat terrain, indicating that the model considers the overall deformation pattern rather than localized features.

Second, the temporal absolute SHAP heatmaps reveal how different time steps influence the classification probability, where certain time intervals exhibit stronger contributions to a specific category. According to temporal heatmaps, the SHAP values of landslides and mining are quite similar, but also differ significantly from subsidence and DSGSD. On one hand, this pattern helps with the distinction between landslide/mining and subsidence/DSGSD. On the other hand, this similarity makes it difficult for the model to distinguish between landslide and mining. And the values for DSGSD are quite low, indicating a lack of a clear temporal pattern, which complicates the classification of DSGSD based on TS alone. The visualization results further confirm that the proposed DL architecture effectively uses the spatial and temporal features to distinguish different types of ground deformation.

E. Discussion

Our experimental results confirm that the proposed multimodal DL model is effective for classifying ground deformation types. In the TS processing branch, we adopted a hybrid 1D-CNN–LSTM architecture, informed by both literature [49] and extensive empirical testing. As shown in Table V,

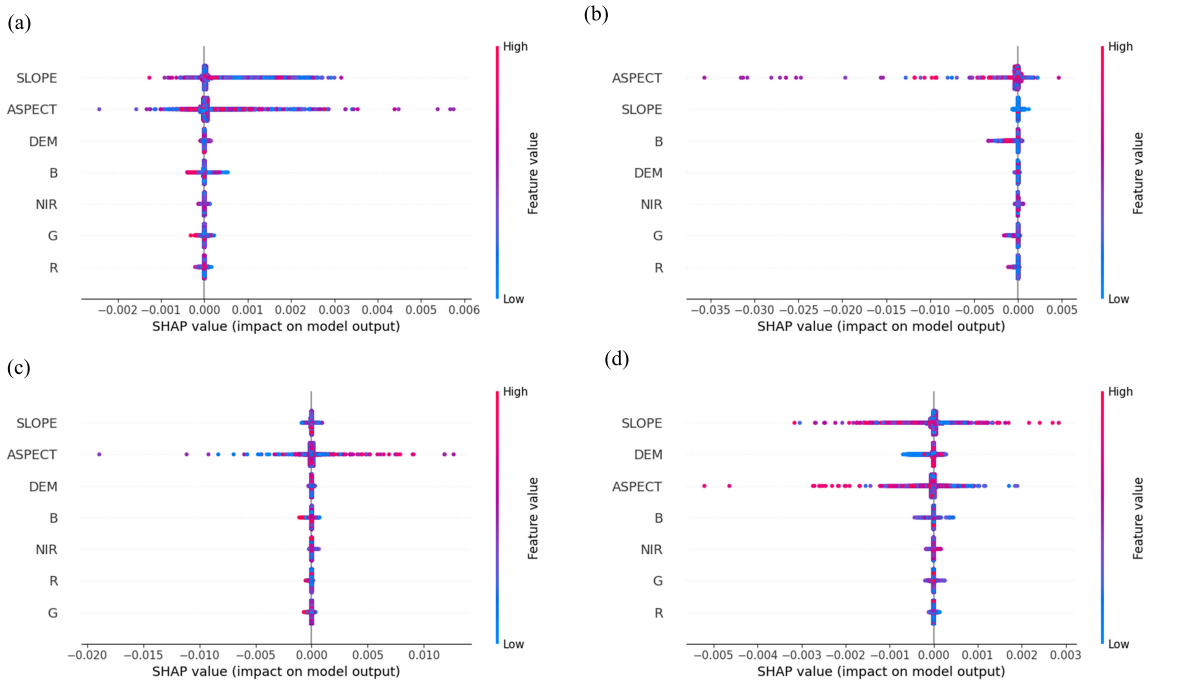


Fig. 8. SHAP summary plot of the image branch for 100 correctly classified samples per class. The X-axis shows SHAP values for feature, dots' density reflects the distribution of SHAP values, and color denotes the original feature values. (a) Landslide. (b) Mining. (c) Subsidence. (d) DSGSD.

TABLE V
PERFORMANCE COMPARISON OF LSTM-BASED MODELS WITH DIFFERENT CNN CONFIGURATIONS ON THE ASCENDING DATASET

Model	Precision (%)	Recall (%)	mF1 (%)	OA (%)
LSTM	51.77 (±7.52)	42.35 (0.54)	39.73 (0.74)	69.75 (±0.52)
1-Layer CNN + LSTM	66.97 (±2.80)	53.03 (±1.24)	54.79 (±1.65)	73.93 (±0.56)
3-Layer CNN + LSTM	**68.35** (±1.51)	**58.97** (±2.97)	**61.44** (±3.28)	**77.56** (±0.64)

this hybrid approach consistently outperforms the standalone LSTM model, with the three-layer CNN + LSTM configuration achieving the best results across all the evaluation metrics. The CNN layers help extract local patterns and compress the input into more compact, informative representations, while the LSTM captures long-term dependencies. By combining the strengths of both the modules, the hybrid architecture offers a more comprehensive understanding of TS characteristics.

We also explored multiple strategies for feature-level fusion of spatial and temporal data, including plain concatenation, attention mechanisms, and LSTM-based fusion. Among them, the LSTM-based fusion module yielded the best performance. A likely explanation is that LSTM has the ability to model temporal dependencies not only within TS data but also across the latent features of the two modalities when treated as sequential inputs. In contrast, attention-based fusion may have underperformed due to insufficient training data, since attention mechanisms typically require large-scale datasets to generalize effectively and avoid overfitting. With additional labeled samples or pretraining strategies, attention-based modules might show greater potential and should be explored in future work.

This study focuses on the integration of multimodal data by proposing an end-to-end framework that combines spatial and temporal information. Rather than reproducing previous ML models, we aim to establish an automated pipeline with minimal manual intervention, which can be adapted to other regions with limited adjustments.

In addition, it is worth noting that we did not apply specific techniques to address the data imbalance through our training. While methods such as data augmentation or class-weighted loss functions were considered, we ultimately chose to preserve the natural distribution of the data to reflect real-world conditions. This choice partly explains the relatively lower $F1$-scores for the minority classes, but also ensures that the results are representative of practical monitoring scenarios.

Several possible extensions of this study could be explored in future work. It remains challenging to distinguish landslide from DSGSD. It should be noted that although both belong to slope deformation processes, they differ in several key aspects. Unlike typical shallow landslide, DSGSD refers to a group of mass-movement phenomena characterized by scarps and cracks but lacks a defined rupture surface and has slower movement rates and large-scale topographic features, while their rheology is different from classical shallow landslides [50]. Therefore, DSGSDs are treated as an independent category to verify whether they can be distinguished. However, DSGSD is commonly observed together with nested landslides, so that motion TS have similar patterns and trends, as well as comparable terrain settings. Moreover, as mentioned earlier, the number of DSGSD samples is smaller than other

classes, the model performs robustly for the majority classes. Another potential direction is to investigate how the spatial patch size used in the image processing branch affects model performance. Different patch dimensions may influence the model's ability to capture relevant spatial features at various scales. Finally, while the current study focuses on classification, integrating deformation forecasting could provide valuable support for hazard assessment and early warning systems. Future research could explore techniques such as data augmentation to address class imbalance [51], conducting sensitivity analyses on patch size to optimize spatial feature extraction, and investigate the integration of TS forecasting models to enhance predictive capabilities [52], and expanding the classification scheme to include additional deformation types for a more comprehensive representation of ground movement processes.

V. Conclusion

In this article, a novel feature-level multimodal data fusion approach has been proposed for ground deformation processes classification by integrating InSAR displacements TS with GIs. Specifically, a dual-branch DL architecture equipped with both the TS and image processing modules is designed to extract representative features from TS and GIs, followed by a LSTM-based fusion module. Extensive results on both the ascending and descending datasets have confirmed the effectiveness of our data fusion strategy. One of the key innovations of the proposed approach is the integration of temporal and spatial features, which significantly improves classification performance compared with single-modal input models.

Furthermore, the large-scale deployment highlights the model's ability to effectively classify diverse deformation types in real-world scenarios. Therefore, the proposed approach can serve as a valuable tool for nationwide ground motion monitoring, supporting geohazard assessment and land-use planning. Based on the results of the XAI analysis, we reveal the different influences of TS and spatial data on the model's decision-making process.

This proposed model could serve as a valuable tool for supporting large-scale ground motion monitoring and geohazard assessment.

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